PD CHECKLIST ASSIGNMENT EVALUATION

Comparison of Healthcare Service and Patient Data Visualization Analytics System

Reference Format: PD Evaluation Rubric – 8 assessment criteria, graded on a 1–4 scale, with a maximum of 32 marks.

Assignment Evaluated: Healthcare Service and Patient Data Visualization Analytics System

Overall Assessment: Fair

S.No.	Assessment Criterion	Comparison / Evaluation	Grade	Auditor Comments
1	Problem Understanding & Formulation	The report explains the healthcare data problem, dataset characteristics, requirements, assumptions and constraints. However, the formulation is somewhat lengthy and could be more focused.	3/4	Good understanding of the problem, with scope for a more concise formulation.
2	Application of Course/Domain Knowledge	The report applies data visualization concepts, variable classification, correlation, PCA and t-SNE. Some explanations are descriptive rather than deeply analytical.	2/4	Relevant knowledge is applied, but the depth of application can be improved.
3	Methodology / Design / Implementation	A clear workflow is provided covering data loading, inspection, visualization, correlation and dimensionality reduction. However, implementation details are not demonstrated extensively within the report.	2/4	Methodology is present, but implementation evidence could be stronger.
4	Use of Appropriate Modern Tools & Techniques	Python 3.12 and appropriate libraries such as pandas, NumPy, Matplotlib, Seaborn, scikit-learn and SciPy are identified and used.	3/4	Tools are appropriate and relevant to the assignment.
5	Results, Testing & Validation	The report includes visual results, ten test cases and performance measurements. More detailed validation evidence and discussion of limitations could strengthen this section.	2/4	Results are presented, but validation could be more comprehensive.

6	Analysis, Trade-offs & Justification	Several visualization and dimensionality-reduction approaches are compared. The choices are justified, but the analysis could be more directly connected to practical healthcare decisions.	2/4	Basic comparison and justification are available; deeper analysis is recommended.
7	Broader Considerations / Professional Responsibility	The report discusses anonymization, privacy, data integrity and responsible interpretation. Economic, environmental, societal and sustainability considerations are less developed.	2/4	Important ethical aspects are covered, but broader professional considerations need more attention.
8	Technical Documentation & Reflection	References, implementation files and a student reflection are included. The reflection addresses learning, challenges and possible improvements, though it could be more personal and detailed.	3/4	Documentation is adequate with a useful reflection section.

Final Score: 19 / 32

Final Rating: Fair (14–20)

Overall, the assignment covers the major requirements of the PD checklist and demonstrates a reasonable understanding of healthcare data visualization. The report contains problem formulation, methodology, tools, results, comparisons, professional considerations, references and reflection. However, a fair rating is appropriate because several areas would benefit from greater depth, especially implementation evidence, validation, analytical justification and broader professional considerations.

Key Areas for Improvement

- Provide more direct evidence of the actual implementation and working process.
- Strengthen the analytical interpretation of visualization results rather than mainly describing them.
- Include more detailed testing and validation evidence.
- Connect visualization findings more clearly to practical healthcare decisions.
- Expand economic, societal, safety and sustainability considerations.
- Make the student reflection more personal and specific to the work completed.

Auditor Summary

The assignment demonstrates a satisfactory level of understanding and implementation of healthcare data visualization concepts. Most checklist requirements are addressed, but the depth and evidence provided are not consistently strong across all criteria. With improved implementation evidence, validation, analytical discussion and broader professional considerations, the assignment could move towards the Good or Excellent range.